IMEN574 / IDSC574 · PROGRAMMING FOR AX

13

Interpretable ML

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Where We Are

01 Programming with AI

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 Deep Learning and Foundation Models

Lectures 14–22

Outline: Interpretable ML

01 **Interpretation Targets**

Define the question

02 **Feature Effects**

Inspect predictive reliance

03 **Local and Example Evidence**

Validate the explanation

01 · Interpretation Targets

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Interpretation Questions

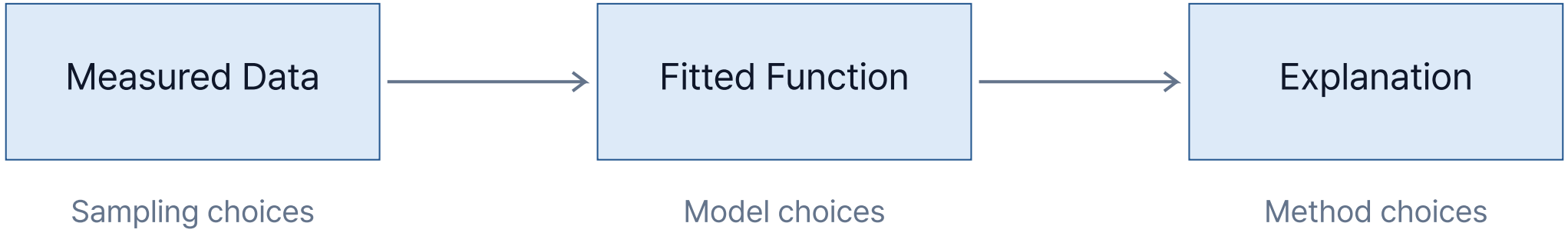
Choose whether to explain a model or one prediction

Scope	Question	Example
Global	What does the model use?	Permutation importance
Local	Why this output?	SHAP, LIME
Action	What could change?	Counterfactual

The question determines which evidence is relevant

Model and World

An explanation describes a fitted function under assumptions



An explanation inherits the model and data limitations

Transparent Model Structure

A simple model exposes its prediction calculation

	Value	Coefficient	Contribution
Feature A	2	2	4
Feature B	4	1	4
Intercept	0	1	0

$$\hat{y} = 2a + b$$

The weighted terms sum to the prediction

Coefficient Scale

Feature units change the numerical coefficient

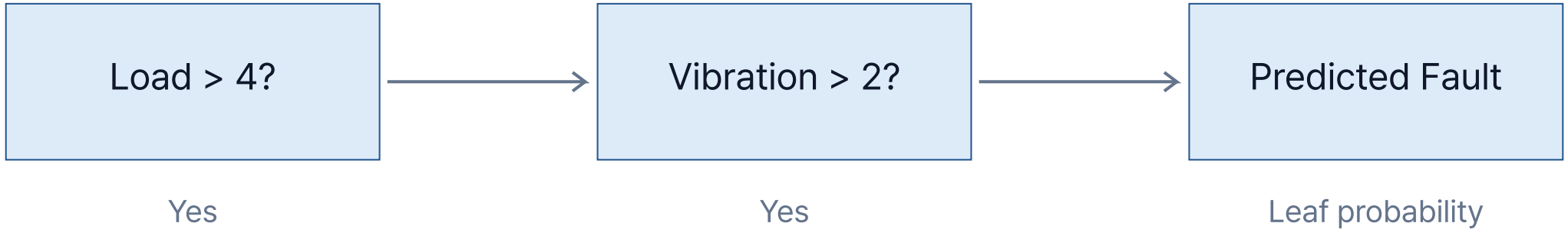
Unit	Coefficient
Meters	2
Centimeters	0.02

$$2 \times 1\ \mathrm{m} = 0.02 \times 100\ \mathrm{cm}$$

The prediction is unchanged although the coefficient differs

Interpretable Trees

A short decision path exposes the applied conditions



A path names the conditions used for this input

Accuracy and Interpretability

Treat the tradeoff as an empirical question

Candidate	Evaluate	Explain
Linear model	Same held-out split	Coefficients
Small tree	Same held-out split	Decision paths
Ensemble	Same held-out split	Post-hoc methods

Compare actual models before assuming complexity is necessary

Explanation Taxonomy

Methods differ by access and scope

	Global	Local
Model-specific	Coefficients	Tree path
Model-agnostic	Permutation	SHAP or LIME

Model access and explanation scope are separate axes

A Known Test Function

Use a small function to verify explanation arithmetic

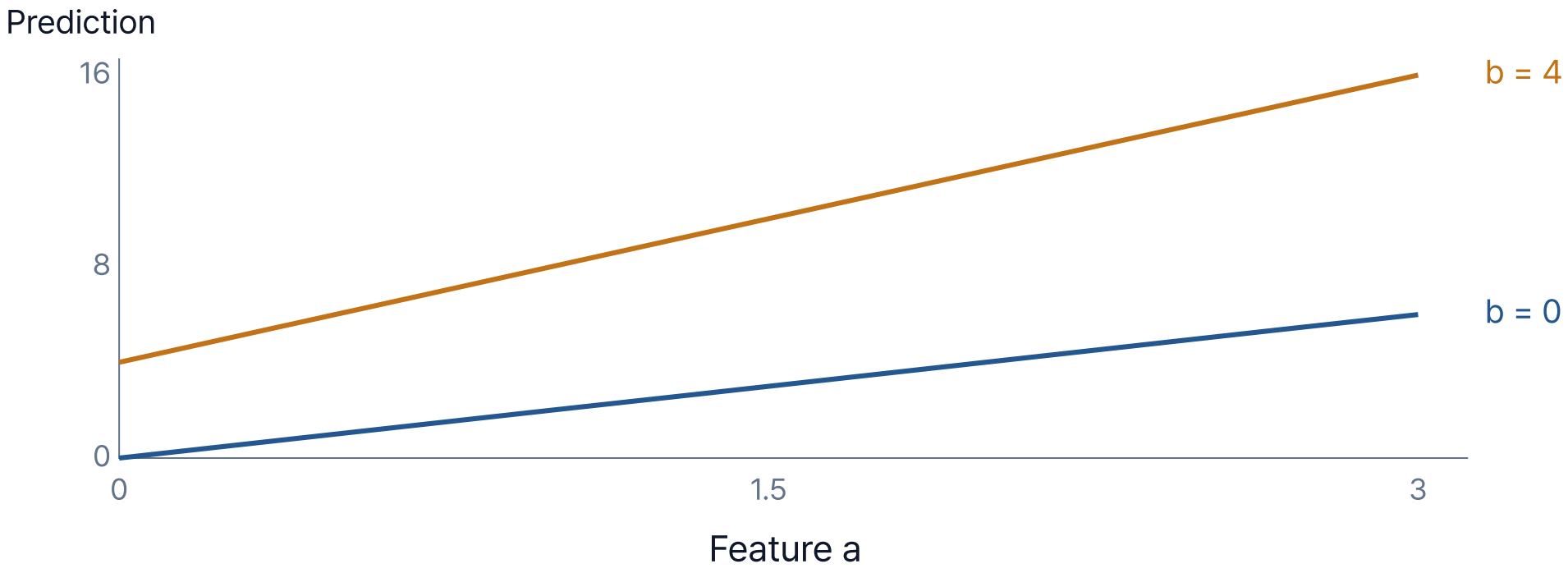
a	b	f(a,b)
2	4	12
1	4	8
2	0	4

$$f(a,b) = 2a + b + 0.5ab$$

The interaction makes feature effects depend on the other value

Feature Interactions

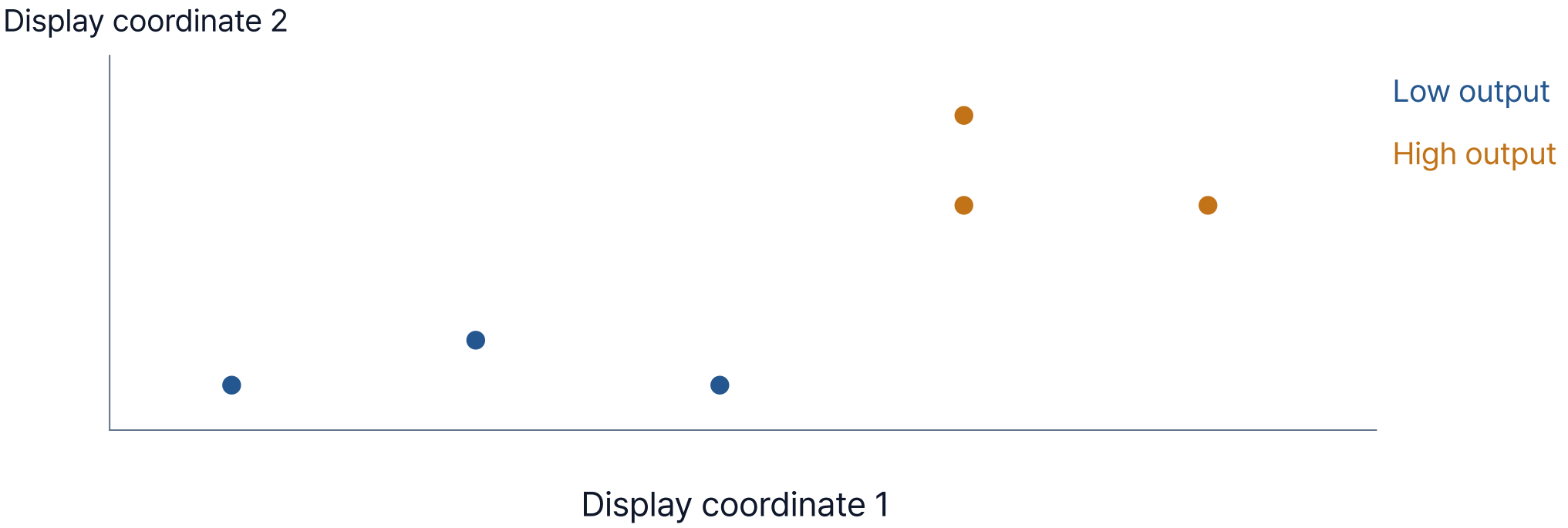
The effect of a changes with b



The slope increases from two to four as b changes

Visualization as Evidence

A projection suggests patterns without explaining the predictor



Colored embedding groups motivate a check rather than a causal claim

Explanation Quality Criteria

Check fidelity, stability, and relevance separately

Criterion	Check
Fidelity	Does the explanation track this model?
Stability	Do small changes alter the result?
Relevance	Does it answer the user question?
Validity	Are perturbed inputs plausible?

A readable explanation can still fail fidelity or validity

02 · Feature Effects

01 Interpretation Targets

Define the question

02 **Feature Effects**

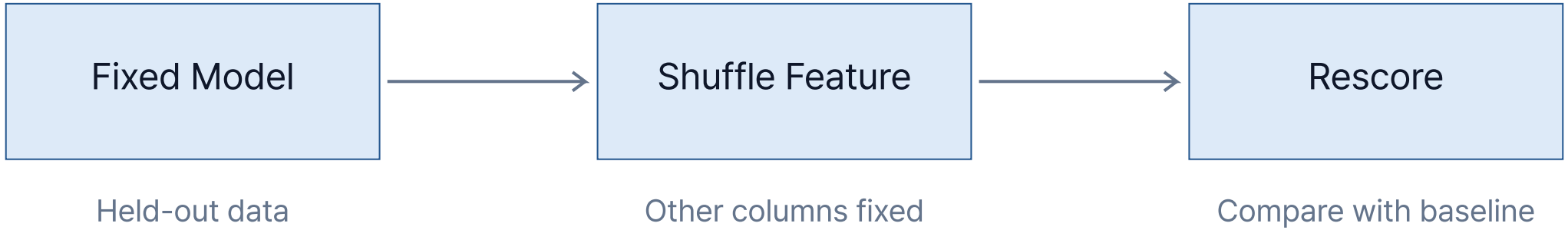
Inspect predictive reliance

03 Local and Example Evidence

Validate the explanation

Permutation Importance

Break one feature relation and measure the score change



The test measures predictive reliance under a chosen perturbation

Permutation Arithmetic

A fixed function makes the loss increase traceable

a,b	True f(a,b)	Prediction after reversing a	Squared error
0,0	0	4	16
1,2	5	5	0
2,4	12	4	64

Reversing a raises MSE from zero to 80/3

Demo: Permutation Loss

Measure how shuffling one feature changes prediction loss

Rotate feature a or b without refitting f Compare shuffled MSE with the baseline loss

Permuted feature a Permutation Reverse

Changed input (a,b)	Prediction	Squared error
2, 0	4	16
1, 2	5	0
0, 4	4	64

Baseline MSE 0; shuffled MSE 26.666667; loss increase 26.666667
The fitted function is fixed; no refitting occurs

Answer: Permutation Loss

The same input produces these calculated values

Input / quantity	Result
baseline a=[0,1,2], b=[0,2,4]	y=[0,5,12]; MSE0
reverse a → [2,1,0]	MSE26.666667
reverse b → [4,2,0]	MSE26.666667

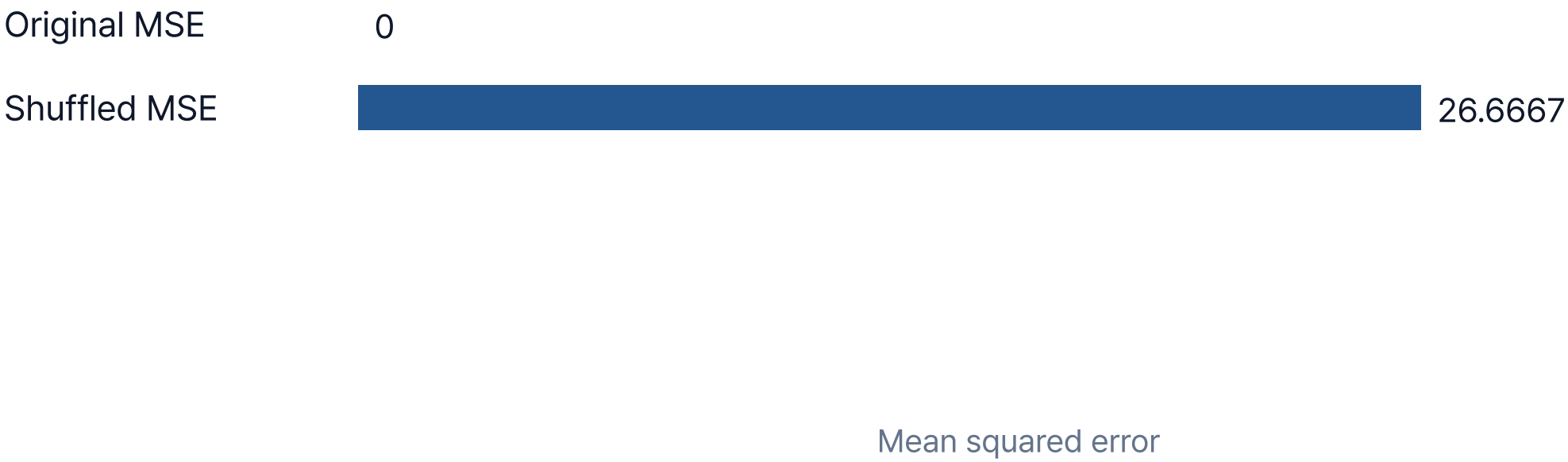
Caveat: Permutation Loss

Permutation importance depends on data and shuffles

Condition	Consequence
Correlated a and b	Permutation creates unsupported combinations
One deterministic shuffle	A worked example, not an uncertainty estimate

A Broken Feature Relation

Reverse a while keeping b and targets fixed



The same fixed targets give a shuffled MSE of 26.67

Caveat: Correlated Predictors

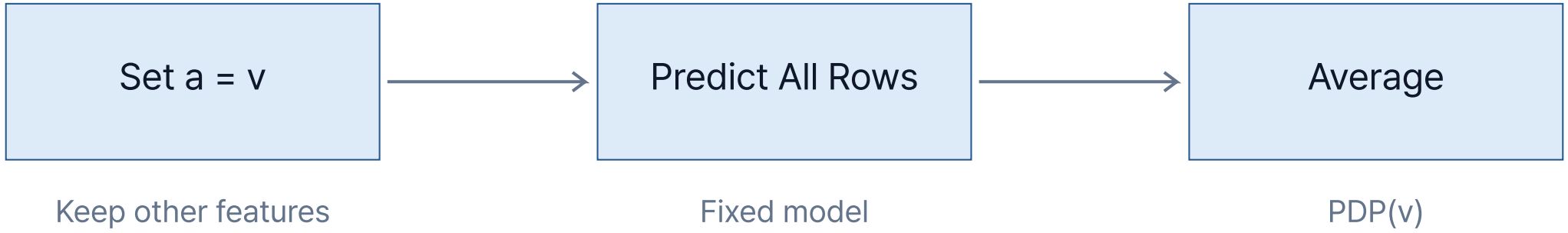
A substitute feature can hide individual reliance

a	Copy of a
1	1
2	2
3	3

Perfectly duplicated columns provide interchangeable evidence

Partial Dependence

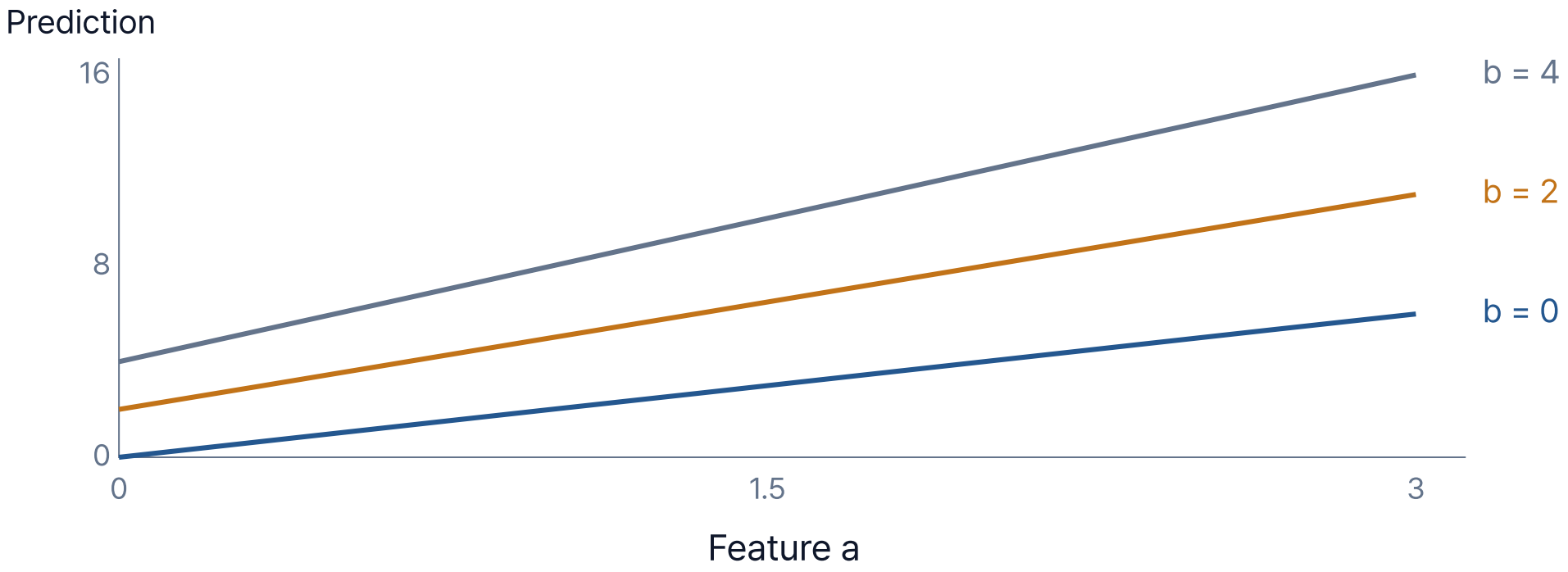
Average predictions after replacing one feature value



The average effect depends on the background rows

Individual Conditional Expectations

Retain separate curves before averaging



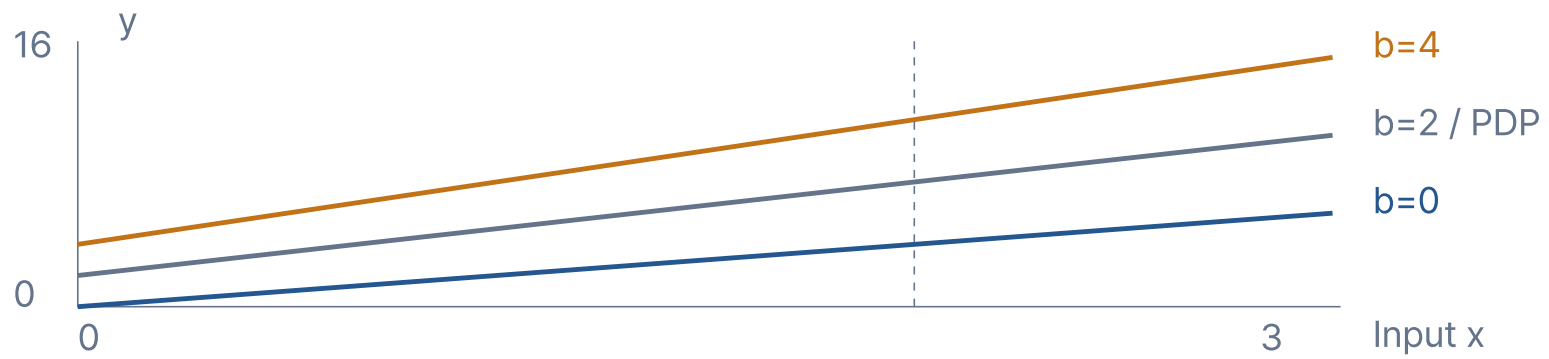
Different slopes reveal interaction with b

Demo: ICE and Partial Dependence

Average individual curves without hiding their differences

Move a while holding each b row fixed Compare the three ICE values with their average

Feature a 2



a=2; ICE [4, 8, 12]; PDP=8

ICE slopes 2,3,4; PDP averages the three fixed b rows

Answer: ICE and Partial Dependence

The same input produces these calculated values

Input / quantity	Result
f(a,b)=2a+b+0.5ab; b=0,2,4	PDP(a)=3a+2
a2	ICE [4,8,12]; PDP8
a1	ICE [2,5,8]; PDP5

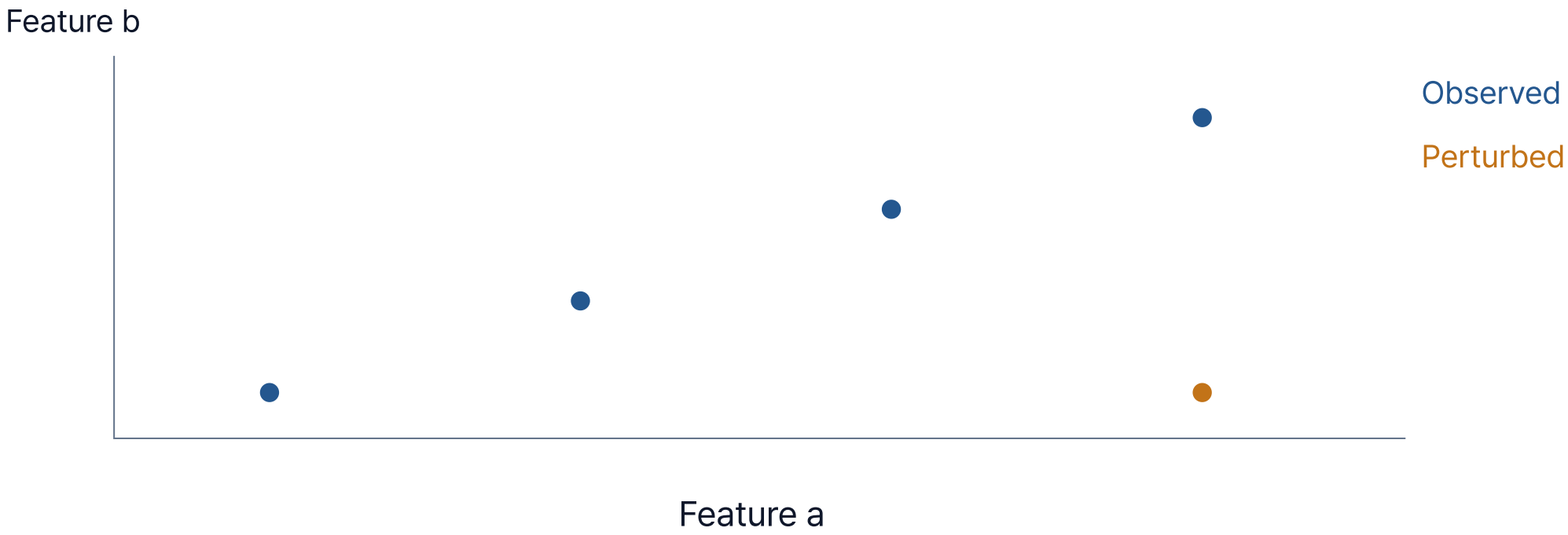
Caveat: ICE and Partial Dependence

Marginal averaging can leave the observed data support

Condition	Consequence
Observed relation b=2a	Setting a while preserving every b breaks the relation

Caveat: Unsupported Combinations

Independent perturbation can leave the observed data manifold



The perturbed point is far from the observed relationship

Shapley Attribution

Average a feature contribution over feature orderings

Ordering	a Contribution	b Contribution
a then b	4	8
b then a	8	4
Average	6	6

The two-feature interaction is shared across both orderings

Additive Explanation Check

Attributions must add to the output relative to a baseline

	Baseline	φa	φb	Prediction
Known Function	0	6	6	12

$$f(x) = \text{baseline} + \Sigma_j \, \varphi_j$$

The contributions sum exactly to twelve in this example

Baseline Dependence

Change the reference population and change the question

Background Point	Model Output
Reference (0,0)	0
Reference (1,1)	3.5

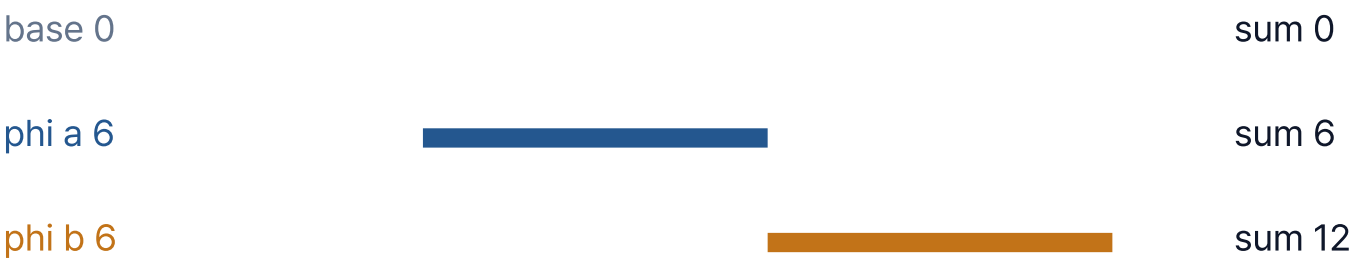
The output difference depends on the reference state

Demo: Exact Two-Feature Attribution

Average both feature orders for an exact two-feature game

Change the reference from (0,0) to (1,1) Check base + phi_a + phi_b = f(2,4)

Reference a 0 Reference b 0



Base 0 + phi_a 6 + phi_b 6 = 12
Exact average over both feature orders for one fixed-reference game

Answer: Exact Two-Feature Attribution

The same input produces these calculated values

Input / quantity	Result
reference (0,0)	base0; phi_a6; phi_b6; output12
reference (1,1)	base3.5; phi_a3.25; phi_b5.25; output12

Caveat: Exact Two-Feature Attribution

Attribution depends on how absent features are defined

Condition	Consequence
Single fixed reference	Exact values for this reference game
Different reference or background distribution	Contributions can change with the same prediction

Local Attribution Magnitudes

A local magnitude describes one selected input



Attribution magnitude at the chosen input

Equal magnitude does not establish equal causal importance

03 · Local and Example Evidence

01 Interpretation Targets

Define the question

02 Feature Effects

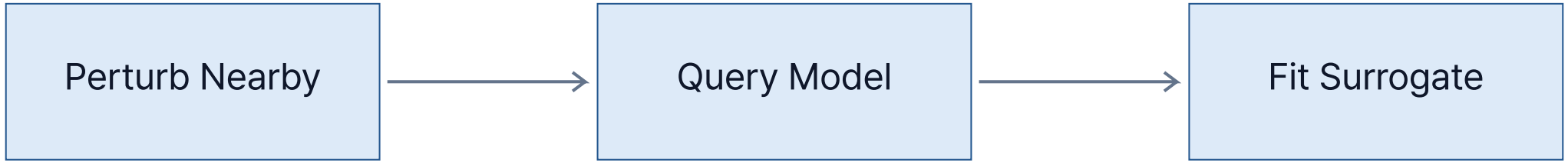
Inspect predictive reliance

03 **Local and Example Evidence**

Validate the explanation

LIME Surrogate

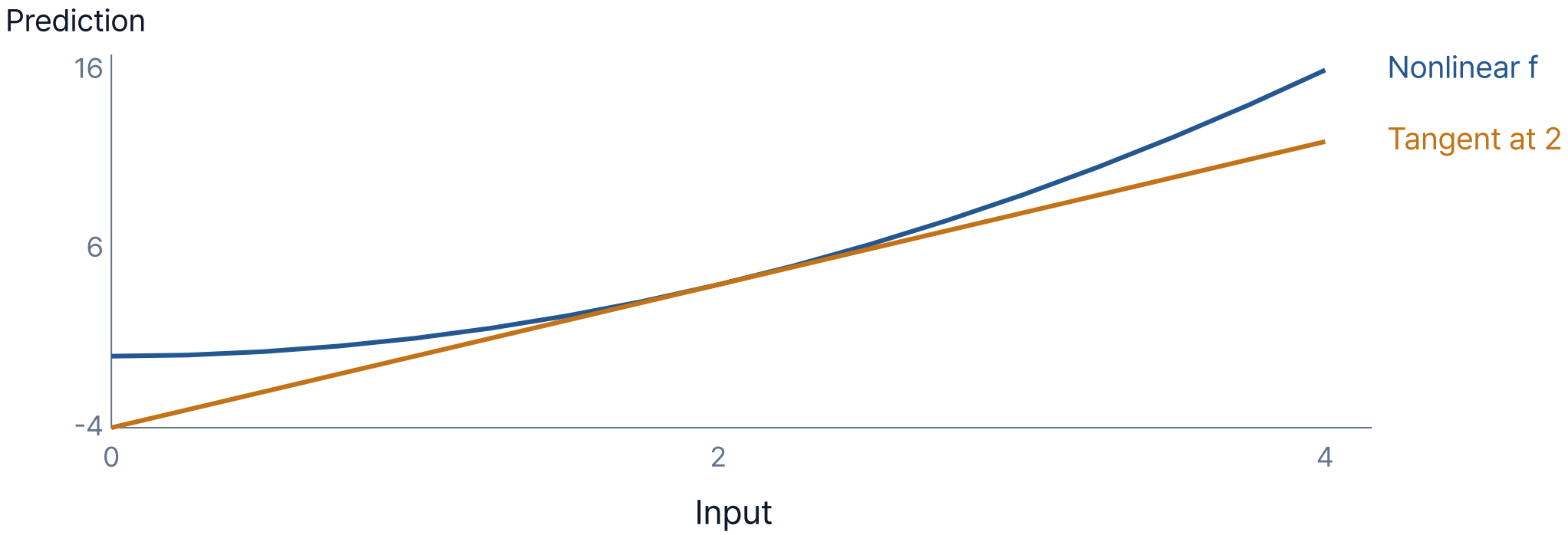
Fit a simple local approximation around one input



A local model approximates the original near one selected input

Local Fidelity

An approximation can be accurate nearby and poor far away



The tangent agrees locally but separates away from two

SHAP and LIME Questions

Choose attribution or local approximation deliberately

Method	Primary Construction	Check
SHAP	Coalition contribution	Background and assumptions
LIME	Local surrogate	Neighborhood fidelity

Different constructions can provide different explanations

Counterfactual Explanations

Search for a plausible input that changes the decision

State	a	b	Prediction
Current	2	4	12
Changed	1	4	8

Reducing a changes the constructed model output from twelve to eight

Feasible Counterfactuals

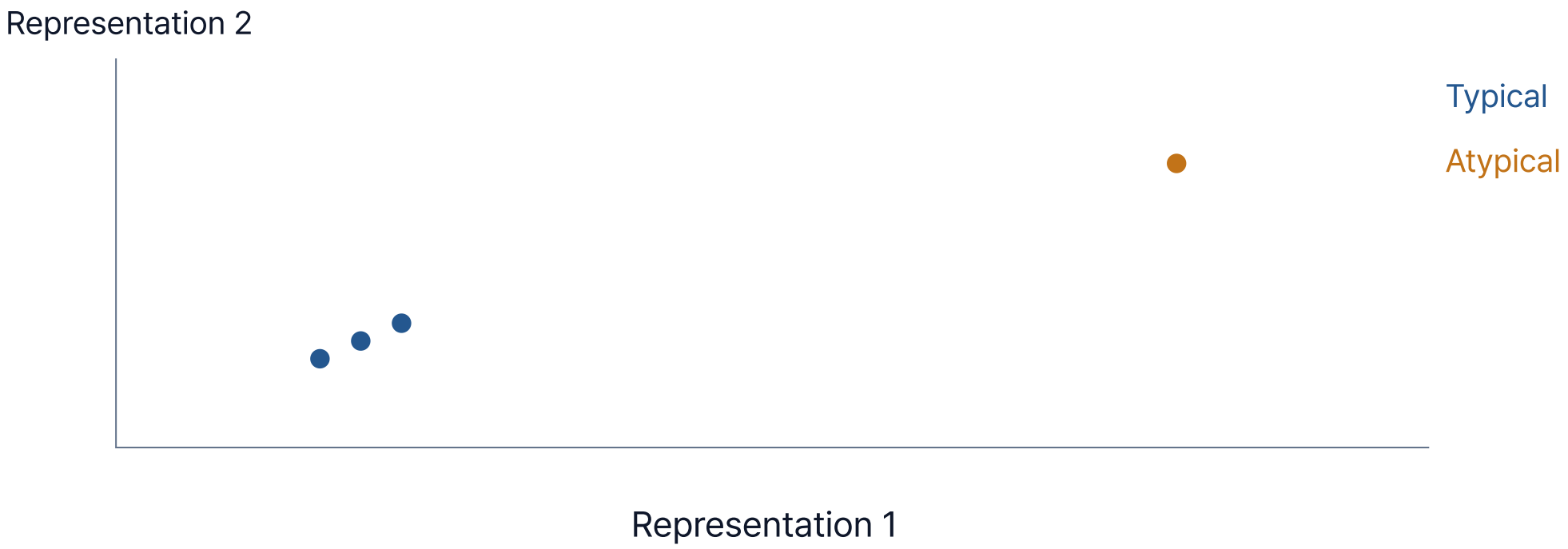
Restrict changes to actionable and valid features

Feature	Constraint
Age	Fixed
Machine setting	Bounded
Unit ID	Fixed
Material mix	Feasible combinations

A mathematical edit may not be a feasible action

Prototypes and Criticisms

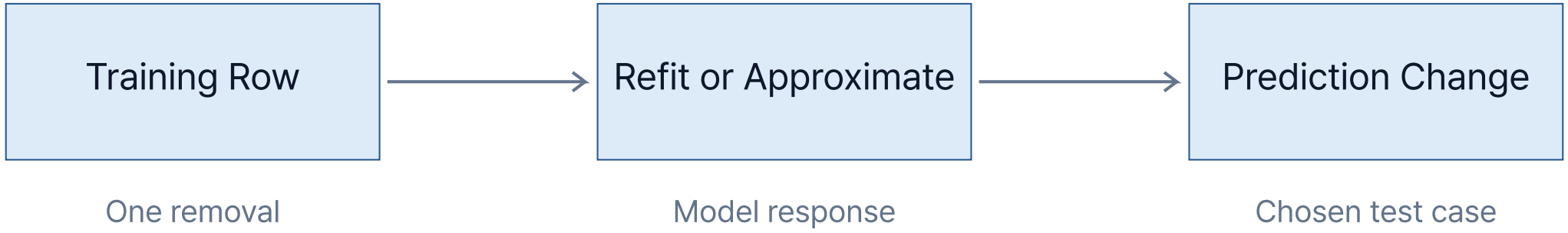
Show representative cases and poorly represented cases



An atypical case exposes what a prototype omits

Training-Example Influence

Ask how a training observation affects the fitted result



Influence concerns fitting rather than feature perturbation

Exact Two-Feature Attributions

Verify the coalition arithmetic without a package

```
def f(a,b): return 2*a+b+.5*a*b
a,b=2.,4.
pa=.5*((f(a,0)-f(0,0))+(f(a,b)-f(0,b)))
pb=.5*((f(0,b)-f(0,0))+(f(a,b)-f(a,0)))
print("a",pa,"b",pb,"sum",pa+pb)
assert np.isclose(pa+pb, f(a,b)-f(0,0))
```

```
a 6.0 b 6.0 sum 12.0
```

Caveat: Explanations and Fairness

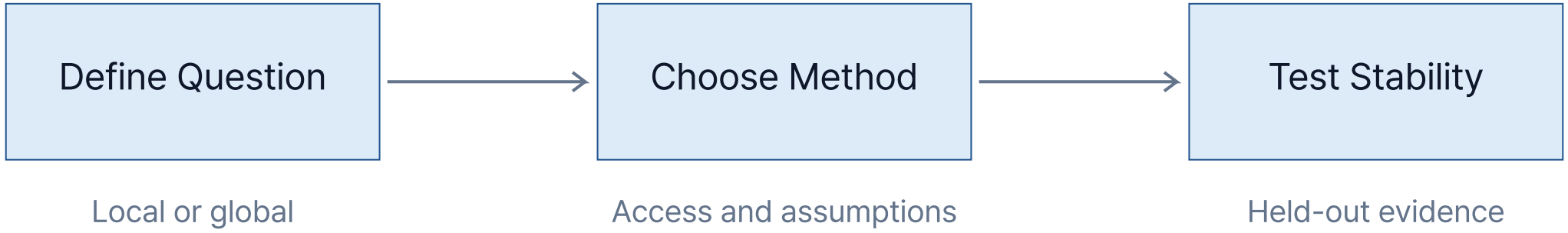
Predictive attribution does not prove fair treatment

Claim	Additional Evidence
Uses a sensitive attribute	Model and input audit
Has disparate errors	Group-specific evaluation
Causes unfair outcomes	Decision process and causal analysis

Different fairness claims require different evidence

An Explanation Audit

Keep the question, model, and perturbation aligned



A trustworthy explanation includes its reference and validity checks

Takeaway: Interpretable ML

01 Explain a chosen model under explicit assumptions

02 Check perturbation validity and reference data

03 Separate explanation from causation and fairness claims